

This document provides the supplementary information about the demonstration videos of the 3DMV algorithm.

1. The videos contain the pedestrian detection results in part of the frames of the three datasets MVD-H, Wildtrack and MultiviewX. To make the size of each video within the 25 MB limit of GitHub, the videos are resized.
2. The video “3DMV_demo_MVD-H.avi” contains the pedestrian detection results at frames 450-499 of the MVD-H Dataset with 4 camera views. The MVD-H video was originally captured at a frame rate of 2 FPS. The demonstration video was generated at a frame rate of 2 FPS. The original resolution of each camera view is 1920×1080 pixels and it was resized to 960×540 pixels in the demonstration video.
3. The video “3DMV_demo_Wildtrack.avi” contains the pedestrian detection results at frames 1800-1995 of the Wildtrack Dataset with 7 camera views. The Wildtrack video was originally captured at a frame rate of 60 FPS. The demonstration video was generated by sampling the original video at a frame rate of 2 FPS. The original resolution of each camera view is 1920×1080 pixels. In the demonstration video, it was resized to 960×540 pixels.
4. The video “3DMV_demo_MultiviewX.avi” contains the pedestrian detection results at frames 360-399 of the MultiviewX Dataset with 6 camera views. The demonstration video was generated at a frame rate of 2 FPS. The original resolution of each camera view is 1920×1080 pixels and it was resized to 960×540 pixels in the demonstration video.
5. The bounding boxes are generated by projecting 3D models of the average pedestrian size to each camera view. For the MVD-H dataset, an octagonal prism model is used, and the bounding box is computed as the minimum rectangle enclosing the projected 16 vertices. For the Wildtrack and MultiviewX datasets, a cuboid model is used, and the bounding box is derived from the projected 8 vertices.
6. Each pedestrian is represented by a distinguished but consistent colour, at a single frame, in all camera views and the synthetic top view. This colour for each pedestrian may change across different frames, since tracking is not included in the 3DMV algorithm.